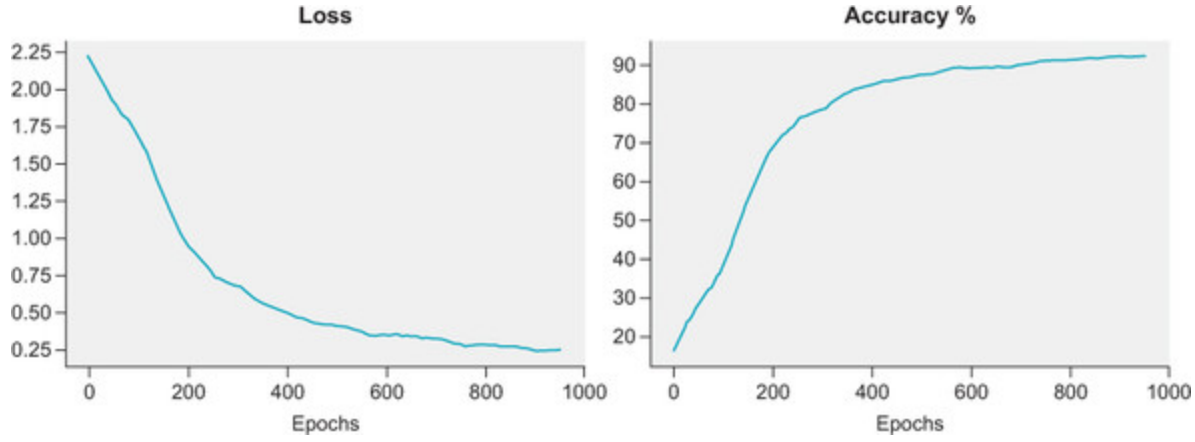
- ***1*** Creates an instance of our relational module
- ***2*** Randomly selects a subset of the MNIST images
- ***3*** Perturbs the images in the batch using the prepare_images function we created, using max rotation of 30 degrees
- ***4*** The predicted image class is the output vector's argmax.
- ***5*** Calculates prediction accuracy within the batch

This is a pretty straightforward training loop to train our MNIST classifier. We omitted the code necessary to store the losses for later visualization, but the unabridged code can be found in this book's GitHub repository. We told the `prepare_images` function to randomly rotate the images by up to 30 degrees in either direction, which is a significant amount.

Figure 10.16 shows how the relational module performed after 1,000 epochs (which is not long enough to reach maximum accuracy). The plots look good, but this is just performance on the training data.

Figure 10.16. The loss and accuracy over training epochs for the relational module on classifying MNIST digits



To really know how well it performs, we need to run the model on the test data, which is a separate set of data that the model has never seen before. We'll run it on 500 examples from the test data to calculate its accuracy.

Listing 10.5. MNIST test accuracy

```
def test_acc(model,batch_size=500):
    acc = 0.
    batch_ids = np.random.randint(0,10000,size=batch_size)
```